IDE.402

CANOPY

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A coffee machine designed for shared environments such as student studios and co-working spaces.

Milestone 1

Empathize

AI: Clarifyer

Coffee Habits | Co-working spaces | Taking a Break | Social Interaction

Selected Impactful Prompt

"Students in co-working studio environments often go to school cafés just before class to get coffee and socialise with friends in a non-academic setting. Because cafés become crowded during this time, students may miss important class information or even attendance. Why could this be happening?"

AI Response Summary

Students use cafés before studio classes as a "Third Space" where they can interact socially without the hierarchy and pressure present in the studio environment. The coffee visit also functions as a pre-class ritual that helps students manage anxiety before entering the high-pressure studio session. However, when many students follow the same ritual, café crowding creates a bottleneck, causing some students to miss the early information and informal guidance shared at the start of class.

Reflections

This suggests that the pre-class coffee moment is not only about consuming coffee but about creating a psychological and social transition between informal peer interaction and the structured studio environment. Therefore, the behaviour indicates a need for a low-pressure social ritual before class, rather than simply a functional need for caffeine.

Design Decisions

Accepted

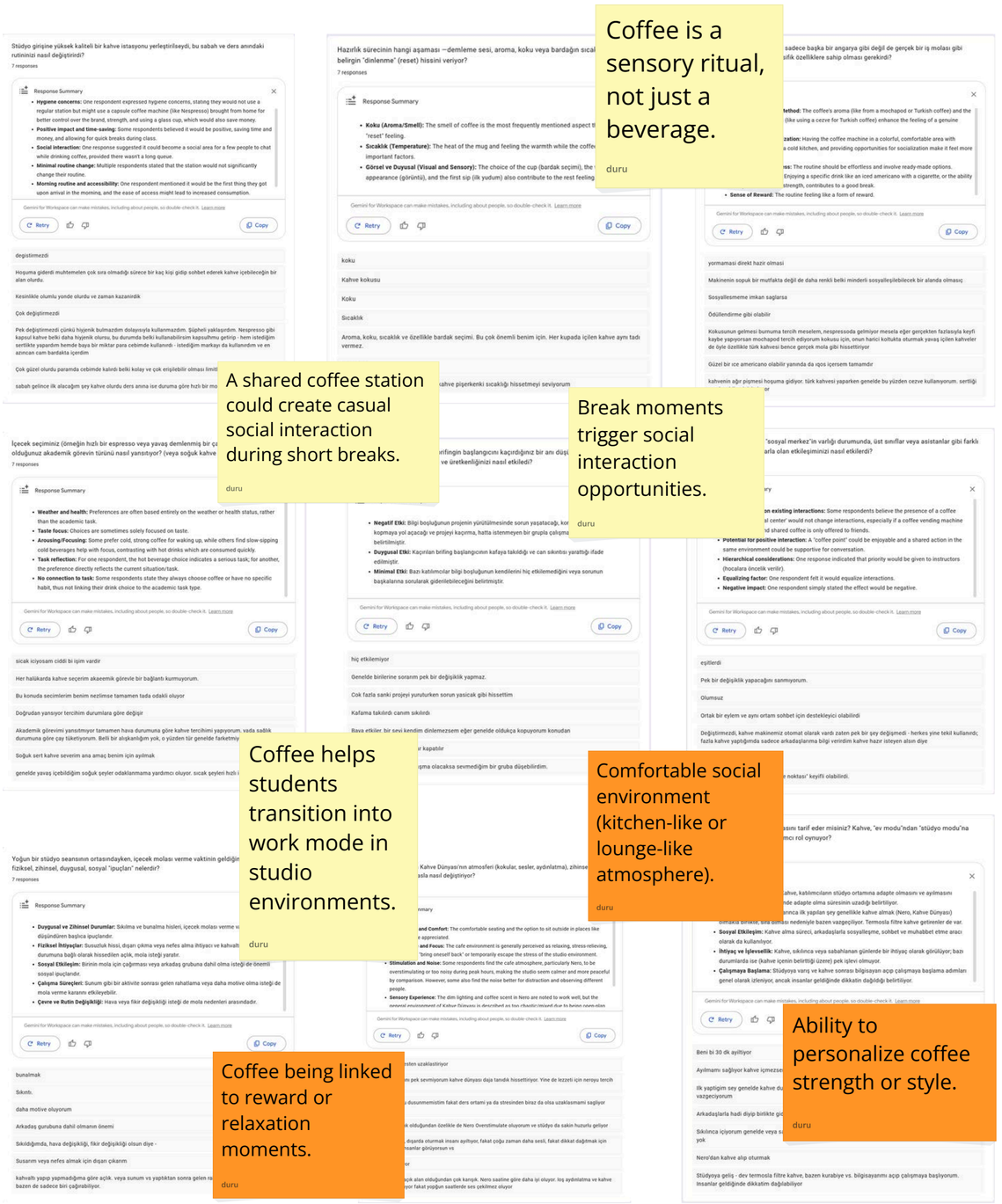
The coffee machine is designed as a shared pre-class ritual located in the studio that allows multiple students to prepare coffee together.

Rejected

Single-user machines and complex digital systems were rejected because they reduce natural social interaction between students

Transformed

The idea of quickly buying coffee was transformed into a social coffee preparation experience that recreates the café "third space" inside the studio.



Milestone 2

Research

AI: Refiner

Optional Co-use | Mental Reset | Third Space | Studio Culture

Selected Impactful Prompt

"Analyse and summarize the results of my open-ended survey about students' coffee habits in studio environments. Understand why students leave the studio to get coffee and what role coffee plays in their daily workflow and social interactions."

AI Response Summary

The results suggested that coffee is not only a caffeine source but also functions as a psychological buffer and social ritual that helps students transition into the studio work environment. The findings also showed that café spaces act as a low-pressure social zone where students exchange informal knowledge, relax, and mentally reset before working.

Reflections

This helped me realize that the design should not focus only on efficiency but also on creating a ritualistic and sensory experience that helps students transition into work mode. Coffee preparation can therefore become a short moment of mental reset and social interaction within the studio environment.

Design Decisions

Accepted

The coffee machine should support a ritualistic preparation process that allows tactile interaction and a short pause before starting work.

Rejected

A purely automatic push-button machine was rejected because it removes the tactile and experiential aspects that users associate with the coffee ritual.

Transformed

The coffee machine concept was transformed into a half-manual, half-automated system that combines convenience with tactile interaction and social presence.

Milestone 3

Ideate

AI: Expander

Form Exploration | Form Variation | Color Exploration | Visual Prompting

Process

I explored how my initial sketch of a two-user capsule coffee machine could be refined while preserving its proportions and product design language. The goal was to use AI to generate form variations and explore alternative interpretations of the concept.

AI Outcomes

Using Vizcom, the sketch was transformed into multiple formal variations that explored different silhouettes, curvature levels, and structural proportions. The tool also generated color and form variations, allowing me to quickly visualize different aesthetic directions for the product.

Reflections

Although the tool successfully generated variations, some results appeared too exaggerated or "monster-like," which indicated that the AI interpretation was strongly influenced by the clarity of the input sketch. I realized that visual prompts and sketches work better than text prompts for my workflow, but they still require careful control and refinement.

Design Decisions

Accepted

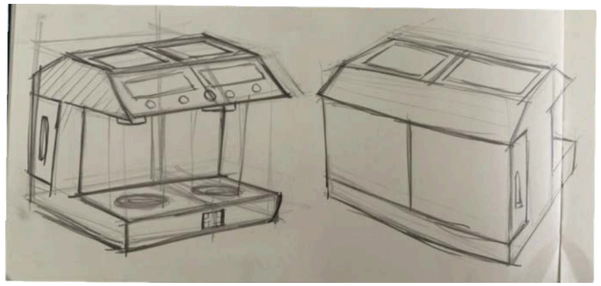
AI was used primarily as a visual ideation tool to explore form and color variations quickly during early concept exploration.

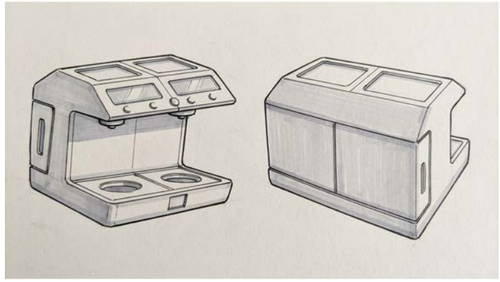
Rejected

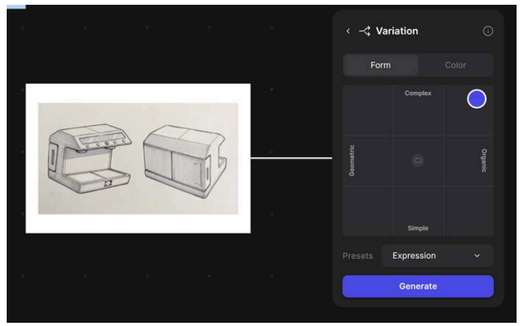
Fully relying on AI-generated forms was rejected because some results distorted the original design intention and proportions.

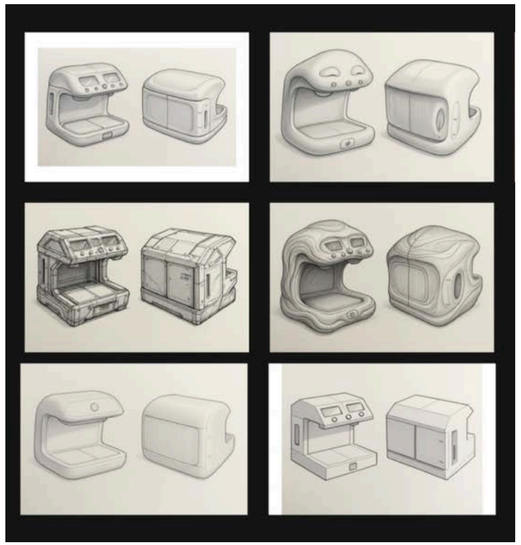
Transformed

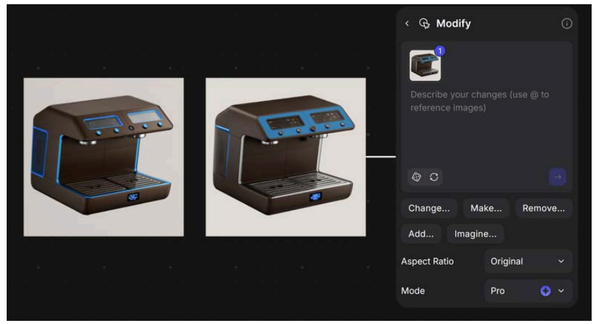
The AI-generated forms were treated as inspiration references rather than final solutions, and the process shifted toward manually refining the geometry by drawing.

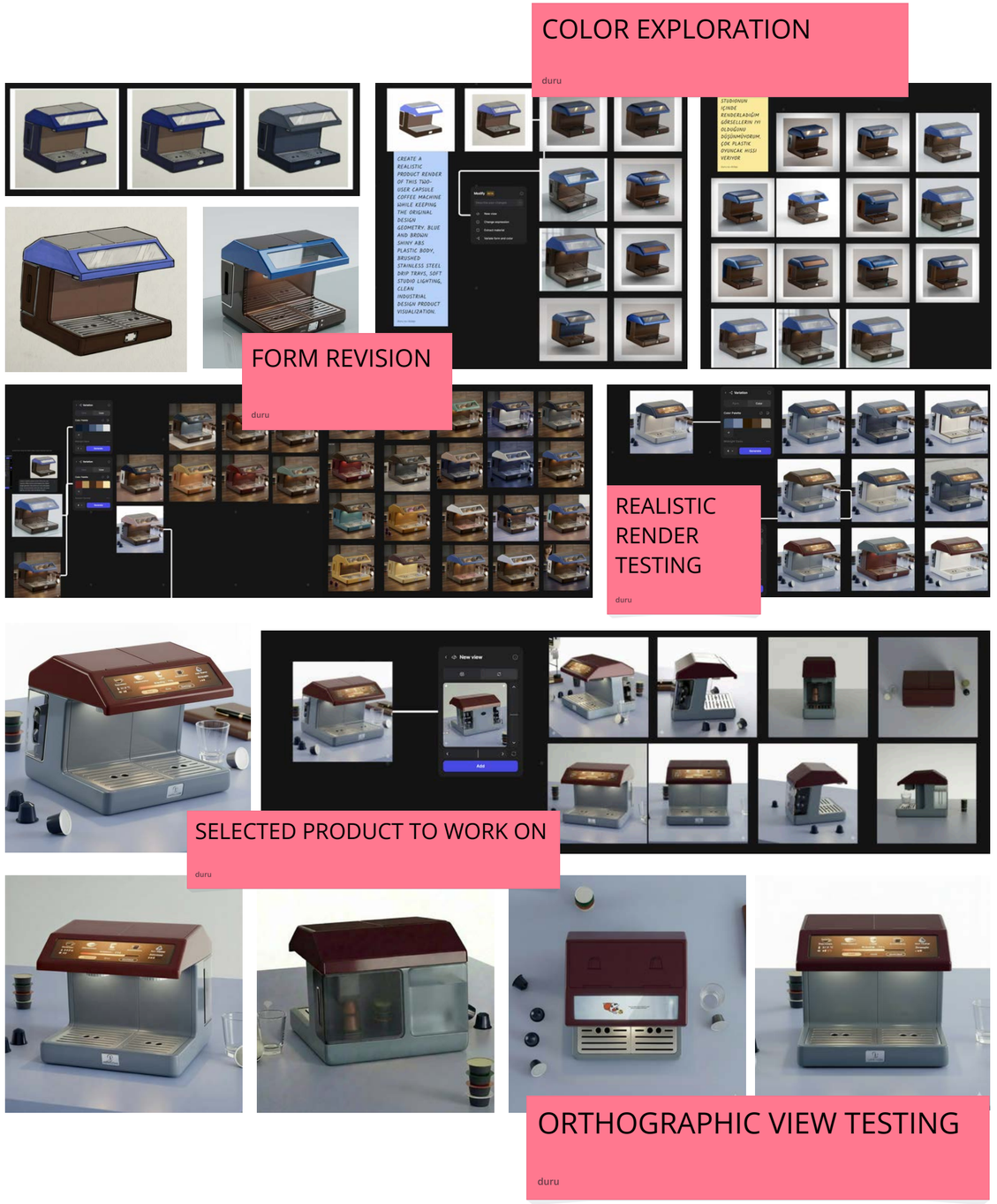












Milestone 4

Ideate / Prototype

AI: Expander

Form Exploration | Form Variation | Color Exploration | Visual Prompting

Process

I used AI tools to generate realistic product renders from my sketch in order to evaluate how the form, materials, and proportions would appear in a more realistic context. The prompt asked the system to create a render of the two-user capsule coffee machine while preserving the original geometry and design structure.

AI Outcomes

The AI generated multiple render variations with different lighting conditions, material interpretations, and color options while keeping the basic form of the machine. However, several outputs introduced distortions, missing components, or unrealistic material expressions, showing inconsistencies in how the AI interpreted the original sketch.

Reflections

The generated results revealed that AI rendering tools can sometimes exaggerate or misinterpret design features, causing the product to appear toy-like or overly artificial. This made me realize that AI visualization is useful for exploring possibilities, but the I must critically evaluate and curate the outputs.

Design Decisions

Accepted

AI-generated renders were used primarily as visual exploration tools to test color palettes, lighting conditions, and material perception.

Rejected

Fully relying on AI-generated outputs for final visualization was rejected because the results occasionally introduced unexpected distortions and unrealistic product interpretations.

Transformed

The workflow was transformed into a comparative process where different AI tools (Vizcom and Gemini) were used to generate and evaluate renders from the same sketch input.

Milestone 5

Define / Prototype

AI: Clarifyer

Design Feasibility | Product Internal Architecture | Product External Revision | Internal Component Layout

Process

To better understand the internal structure of capsule coffee machines, I asked AI tools to generate an exploded view based on my product images and explain the internal components of the system. The goal of this prompt was to visualize how elements such as the pump, thermoblock heater, capsule chamber, and sensors could realistically be arranged inside my design and how it should it be revised outside according to inside.

AI Outcomes

The AI generated a rough exploded view and provided a detailed explanation of the main internal components typically found in capsule coffee machines, including the water tank, flow meter, high-pressure pump, thermoblock heater, brewing chamber, and electronic control board. Although the visual output itself was not entirely accurate, the explanation helped clarify the functional relationships between components and the overall system flow.

Reflections

Through this process, I realized that understanding the internal architecture of existing capsule machines is crucial for designing a believable and technically coherent product. Even though the AI-generated visualization was imperfect, the detailed breakdown of the brewing system helped me conceptualize how the internal structure of my coffee machine could be organized.

Design Decisions

Accepted

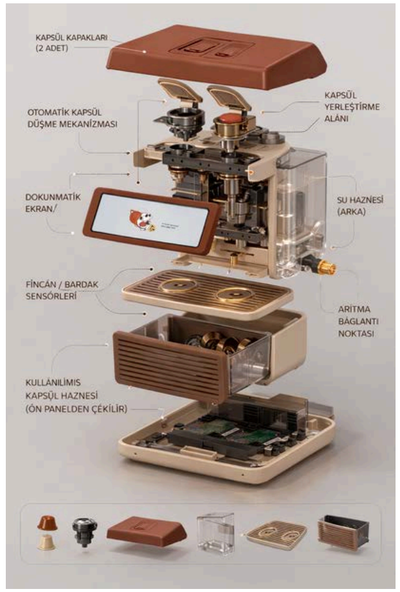
The internal system logic of capsule coffee machines adopted as the technical foundation for my design.

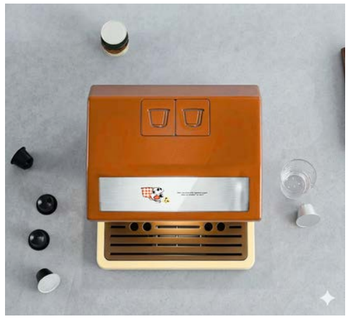
Rejected

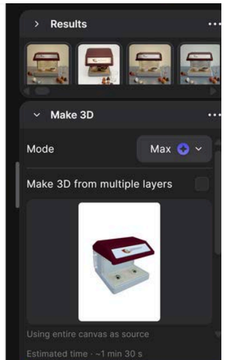
The initial exploded view generated by AI was not used directly because the spatial arrangement of the components was inaccurate and did not correctly reflect the mechanical and formal structure of the machine.

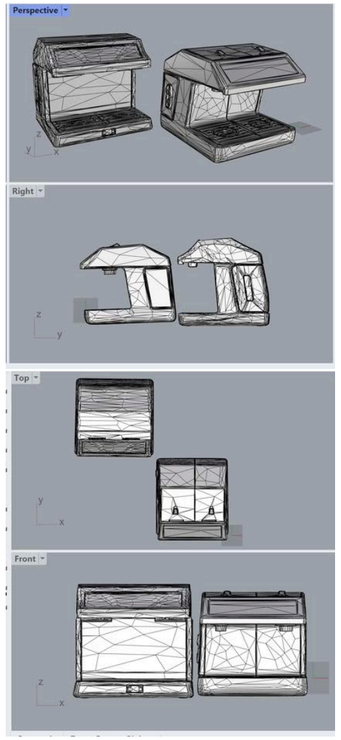
Transformed

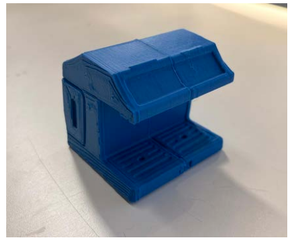
Instead of relying on the AI visualization, I used the information it provided to research real capsule machines (as Nespresso systems) and reinterpret the component layout according to the needs of my own design.











Milestone 6

Prototype/UI Testing

AI: Critic

Social Interaction | BehaviourTesting | UI Prototype

Selected Impactful Prompt

"Will users understand the reward logic of the coffee machine interface and choose to prepare coffee with another person in order to receive the bonus capsule?"

AI Response Summary

The AI analysis highlighted several behavioural risks related to digital trust and usability. It suggested that users might ignore the social feature if the reward system is not immediately clear, choosing the individual coffee option instead.

Reflections

Through prototyping and testing, I observed that participants were curious about the reward system and several users naturally chose the shared coffee option. This showed that when the incentive is visible and simple, the interface can successfully trigger social interaction between users.

Design Decisions

Accepted

A touchscreen interface guiding users through the brewing process was kept as the main interaction method, clearly presenting the option to prepare coffee alone or with a friend.

Rejected

The idea of displaying the reward immediately before the brewing process was rejected because it could shift users' focus toward the reward rather than the shared experience.

Transformed

The reward was integrated at the end of the process through a QR code system, allowing the social interaction to occur naturally during coffee preparation while still providing a bonus incentive afterward.

Milestone 7

Define

AI: Clarifyer

Reward System

Social Interaction

Selected Impactful Prompt

"How can a coffee machine in a studio environment create a natural social ritual without forcing users to interact?"

AI Outcomes

The AI suggested several behavioral mechanisms such as shared waiting moments, slow brewing rituals, and micro gathering points that naturally create opportunities for interaction. Instead of forcing collaboration, the system focuses on temporal overlap and environmental affordances that allow conversations to emerge organically.

Reflections

I realized that the machine itself should not force people to socialize but should create moments where social interaction can naturally occur. The idea of designing small rituals around coffee preparation helped me rethink the machine as a social facilitator rather than just a functional appliance.

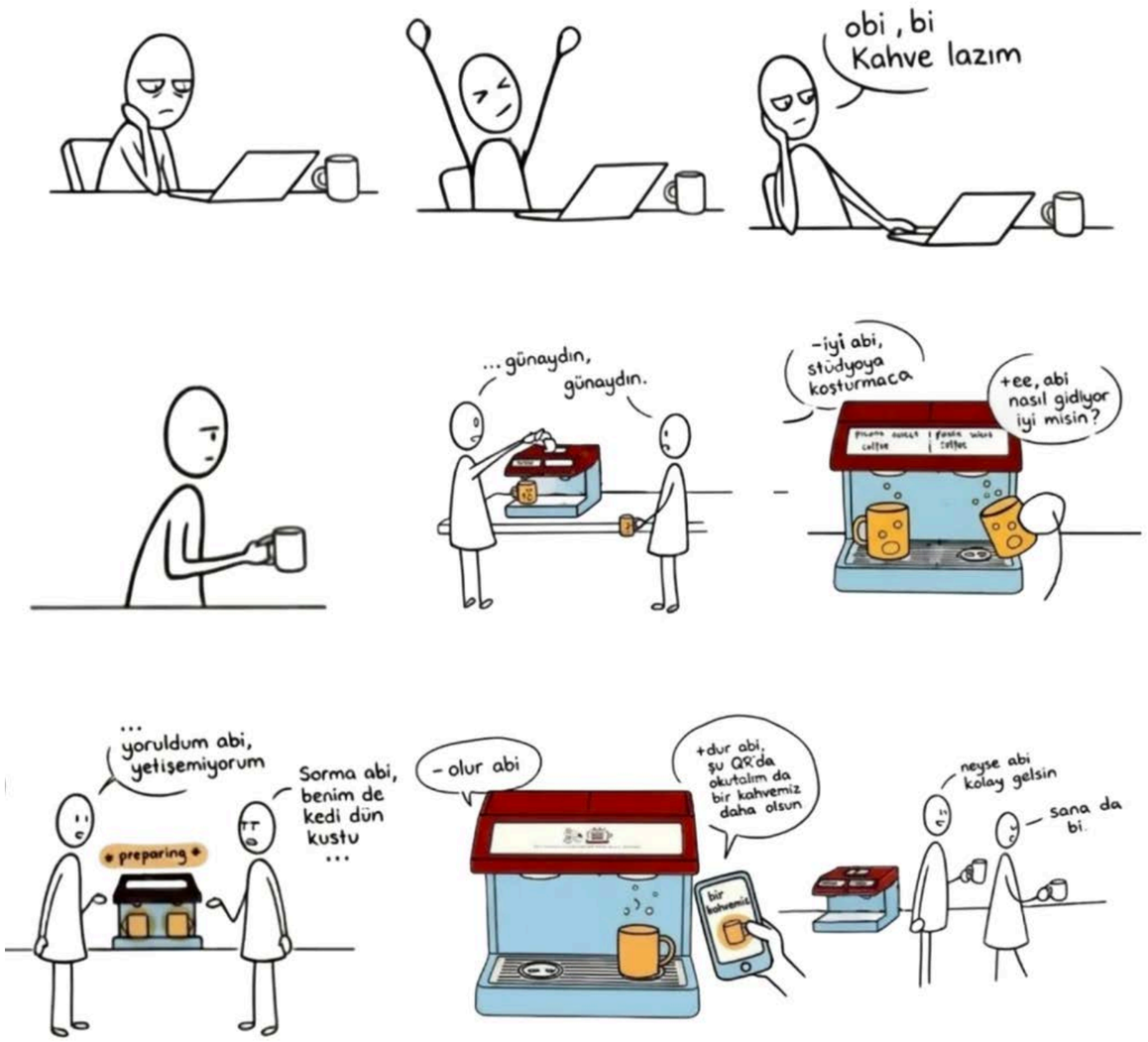
Design Decisions

Accepted

The project adopts the concept of a shared coffee preparation ritual, where short waiting times and visible brewing processes create opportunities for conversation. The machine is designed to support both individual use and spontaneous shared use.

Rejected

A system where the machine only works when two people use it simultaneously was rejected because it would force social behavior and limit usability. Users must always have the option to prepare coffee individually.

Transformed

The initial idea of forcing collaboration was transformed into a “Shared Coffee Pause” ritual, where the machine allows individual use but makes the shared experience more enjoyable and rewarding. This approach creates a natural social micro-break within the studio environment.